RAG-POWERED CAREER & SKILL INTELLIGENCE AGENT

CareerSage AI

A working, tested and deployable product: real inputs, real outputs, every answer cited.

Live product demo | 22 Oct 2026

WHY CAREERSAGE

Problem > Solution

The Problem

Career advice is generic and outdated.

Chat AI invents skills, courses and requirements.

No way to verify where an answer came from.

The Approach

Retrieval-Augmented Generation: answer only from a curated knowledge base.

Out-of-scope questions are refused, never guessed.

The Outcome

Roadmaps, interview prep, skill-gap score and role matching, each with a [source-id] you can check.

ARCHITECTURE

How It Works

Normalize
Aliases: ML > machine learning

>

Typo-fix
"machin" > machine

>

Scope guard
No role or skill term? Refuse

>

BM25 search
27 docs, 9 roles x 3 types

>

Intent boost
Roadmap / interview / skills

>

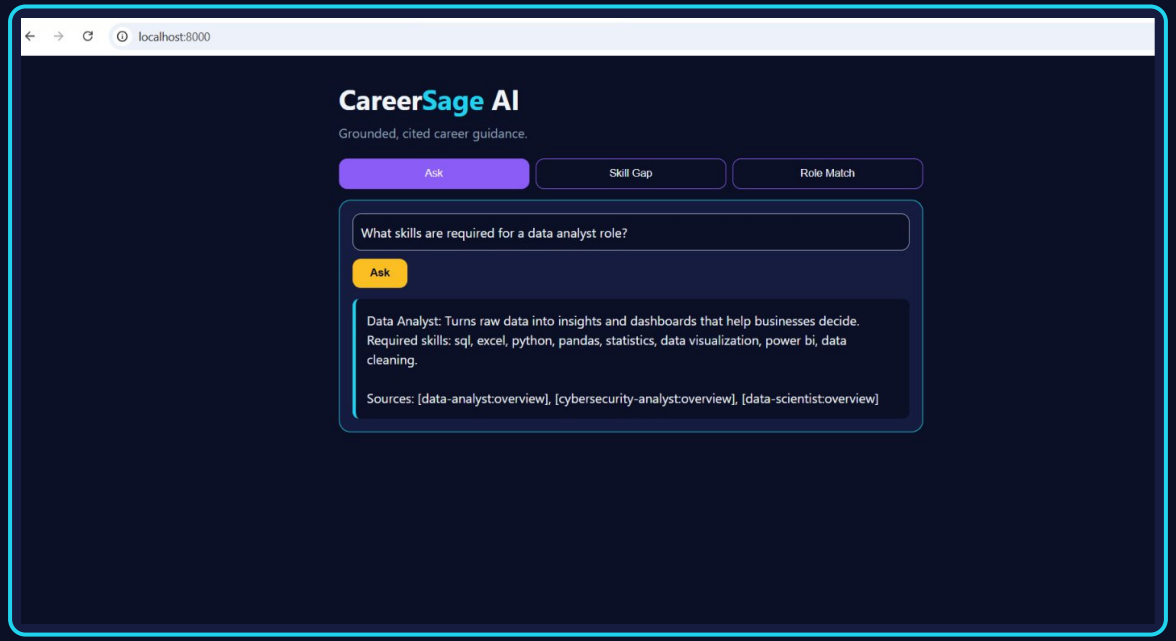
Cited answer
LLM optional, extractive fallback

>

Python standard library only: zero dependencies, so it runs and deploys anywhere

REAL OUTPUT

Live Demo 1: Ask a Career Question



1 Natural question

"What skills are required for a data analyst role?"

2 Grounded answer

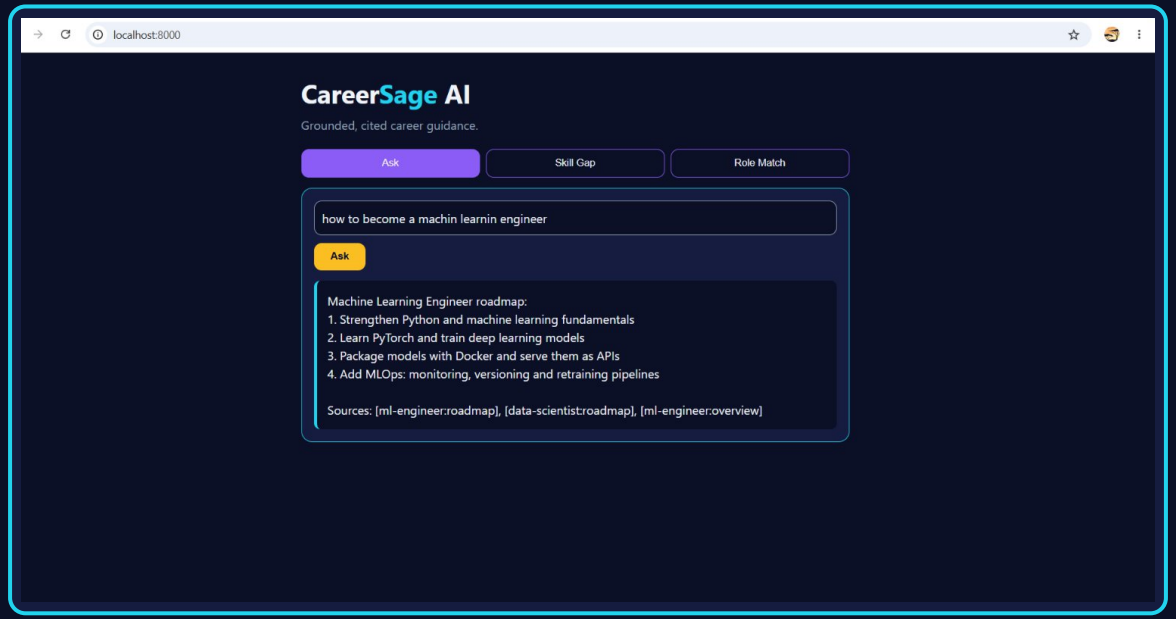
Role summary and exact skill list from the knowledge base

3 Verifiable sources

Every answer ends with [doc-id] citations

REAL OUTPUT

Live Demo 2: Typos and Intent



Typo-tolerant

"machin learnin" was understood as machine learning

Intent detected

"how to become" returns the roadmap, not a summary

Top source correct

[ml-engineer:roadmap] ranked first

PERSONALISED INTELLIGENCE

Skill-Gap Analyzer and Role Match

Skill Gap (Backend Developer)

Resume: Python, SQL, Docker, Git, FastAPI, testing

Result: 75% match
Missing: rest apis, system design
+ 4-step roadmap and interview focus

Role Match

Resume: html css javascript react typescript git

Result: #1 Frontend Developer
Ranked by % of required skills matched

Both examples are asserted in the automated test suite, so the numbers are reproducible.

QUALITY

Testing and Guardrails

23 / 23	100%	100%	0
Automated tests passing	hit@1 on 35-question golden set	Out-of-scope refusal	Third-party dependencies

Metrics are measured on our golden set and gated in CI. Also: input validation, 100 KB body cap, per-IP rate limit, non-root Docker.

DEPLOYMENT

From Laptop to Public URL

1 Run locally

python -m app.server

Open localhost:8000

UI + API in one process

2 Containerize

Docker image, non-root user, built-in health check at /health

3 Ship

GitHub Actions runs tests + eval, then Render deploys from render.yaml to a public link

WHAT NEXT

Roadmap

Retrieval upgrades

Dense embeddings + Qdrant hybrid search
Cross-encoder reranker
Live job-posting ingestion

Product upgrades

Marathi / Hindi queries
User accounts and memory
RAGAS faithfulness scoring in CI

Thank you - Questions?